

TOPIC 3, PART A

Demand and Marginal Personal Worth

Principles of Microeconomics

Prof. Bryan Cutsinger • Florida Atlantic University

People Need Food. But How Much, and at What Price?

We call some things needs: food, housing, medicine, clean water.

The label sounds like it settles the quantity — as if the amount required were fixed, and cost beside the point.

So answer the question the label skips: **how much, and at what price?**

Last Time You Got Five Questions. Today You Get the Tools.

FROM TOPIC 1

What is scarce? How is it allocated?

You already have the five questions.

TODAY

How much will people buy, and what is it worth to them?

Demand and marginal worth.

THURSDAY

How big is the response?

Elasticity, and what it is good for.

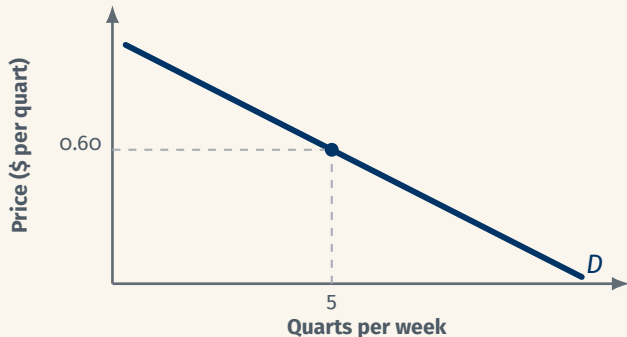
Most of today is problems, not lecture.

01

The Law of Demand

A demand schedule is a menu of alternatives, not a diary of purchases.

A Lower Price Moves This Buyer to a Larger Quantity



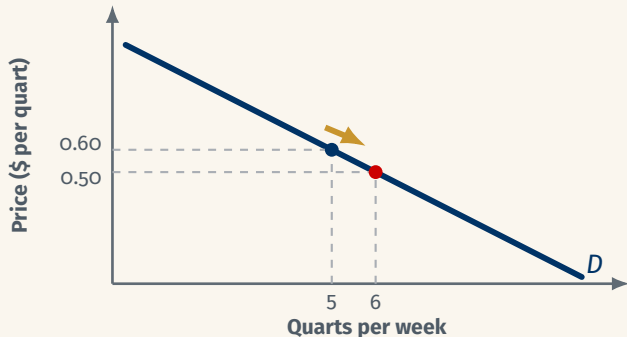
THE SCHEDULE

Price	Quarts/wk
\$0.70	4
\$0.60	5
\$0.50	6
\$0.40	7

Each row is one point on the curve.

At \$0.60 this buyer takes five quarts per week.

A Lower Price Moves This Buyer to a Larger Quantity



THE SCHEDULE

Price	Quarts/wk
\$0.70	4
\$0.60	5
\$0.50	6
\$0.40	7

Each row is one point on the curve.

Same curve, new point: quantity demanded rose from five to six — demand did not change.

“Vacations Got Cheaper and I Still Take One a Year.”

Cheaper vacations, and still one trip a year. Is the law of demand wrong?

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THE TRAP

Counting trips

“Quantity” got read as the number of departures.

THE OPEN MARGINS

Longer. Farther. Nicer.

Ten days instead of five.
Europe instead of Orlando.

THE LESSON

Find the margin before you test the law

The response shows up
wherever it is cheapest to
adjust.

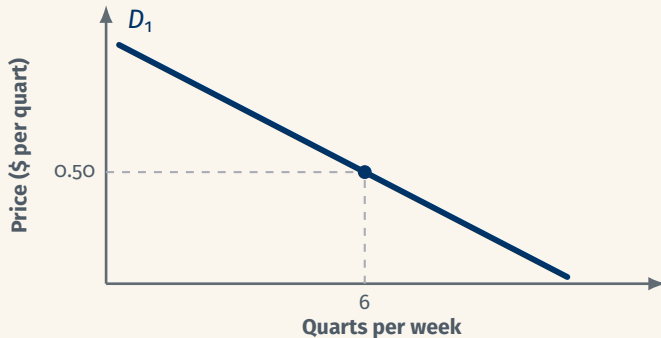
The law held. The quantity of vacation rose — just not in trips.

02

Slide or Shift?

Did the good's own price change, or did a relevant demand condition change?

Own Price Slides the Point. Changed Conditions Shift the Curve.

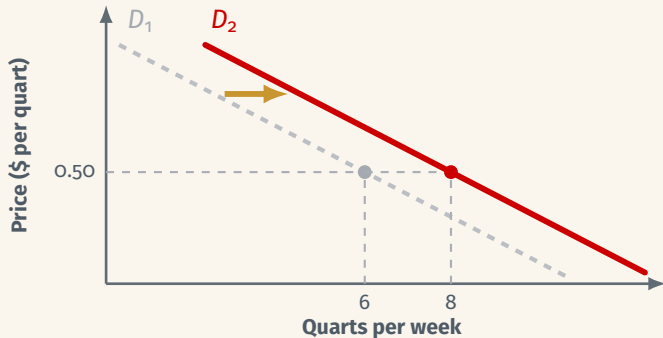


At \$0.50 this buyer chooses six quarts on D_1 .

WHAT SHIFTS THIS BUYER

- 01 Income or wealth
- 02 Prices of related goods
- 03 Tastes
- 04 Expectations

Own Price Slides the Point. Changed Conditions Shift the Curve.

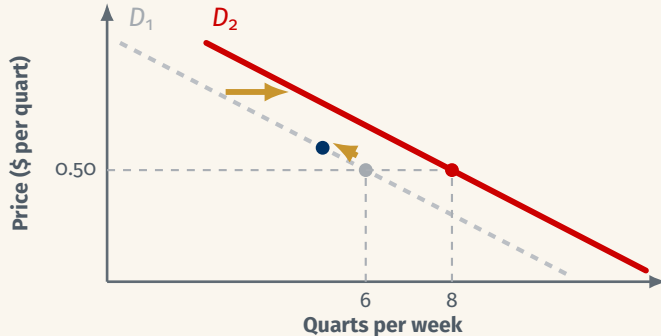


WHAT SHIFTS THIS BUYER

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The shift: at the same \$0.50, quantity rises from six to eight.

Own Price Slides the Point. Changed Conditions Shift the Curve.



WHAT SHIFTS THIS BUYER

- 01 Income or wealth
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- 04 Expectations

Slide or shift: *along D_1 , own price changed; between the curves, a condition changed.*

Seven Wig Scenarios: Slide or Shift?

For each: slide along the curve, or shift the whole curve — and which way?

What happened	Slide or shift?
Your salary goes up	
The price of hats rises	
A new swimming pool opens	
Hair care gets more expensive	
You get divorced	
More people wear wigs	
The price of wigs falls	

Seven Wig Scenarios: Slide or Shift?

For each: slide along the curve, or shift the whole curve — and which way?

What happened	Slide or shift?
Your salary goes up	Shift right — more income
The price of hats rises	Shift right — substitute got costlier
A new swimming pool opens	Shift right — swimming wrecks hairdos
Hair care gets more expensive	Shift right — substitute got costlier
You get divorced	Shift right — back on the market
More people wear wigs	Shift right — tastes moved
The price of wigs falls	Slide — down the same curve

Six shifts, one slide — and only the last one touched the wig's own price.

Beef Gets Pricier. What Happens to Hamburger Buns?

Beef's price rises. Does bun demand shift right, left, or not at all?

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Relationship	Price of A rises...	Example
Substitutes	demand for B rises	beef up → chicken demand right
Complements	demand for B falls	beef up → bun demand left
Independent	no shift	beef up → notebooks unmoved

Buns complement beef, so bun demand shifts left.

03

What One More Unit Is Worth

The height of the curve is the most you would pay for the next unit.

The Next Unit Goes to the Next-Best Remaining Use

FIRST QUART

Highest-valued available use

This buyer would give up as much as \$1.00 for it.

SECOND QUART

Next-highest available use

With the first use already served, this one is worth \$0.90.

SIXTH QUART

Lowest-valued use still chosen at \$0.50

The seventh is worth only \$0.40, so it is left behind.

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Marginal personal worth The most this buyer would give up for one more unit. **The buying rule:** take every unit worth at least the price; stop at the first one that isn't.

TEST IT At \$0.50, why six quarts and not seven?

Worth, Spending, and Surplus All Come from the Schedule

Price	Quarts/wk
\$1.00	1
0.90	2
0.80	3
0.70	4
0.60	5
0.50	6
0.40	7
0.30	8
0.20	9
0.10	10

The milk schedule from the post, all ten rows. Everything else on this slide is one arithmetic step away.

Worth, Spending, and Surplus All Come from the Schedule

Price	Quarts/wk	Marginal worth
\$1.00	1	\$1.00
0.90	2	0.90
0.80	3	0.80
0.70	4	0.70
0.60	5	0.60
0.50	6	0.50
0.40	7	0.40
0.30	8	0.30
0.20	9	0.20
0.10	10	0.10

Marginal worth reads straight off the schedule: the price at which she just takes that unit is what it is worth to her.

Worth, Spending, and Surplus All Come from the Schedule

Price	Quarts/wk	Marginal worth	Total worth
\$1.00	1	\$1.00	\$1.00
0.90	2	0.90	1.90
0.80	3	0.80	2.70
0.70	4	0.70	3.40
0.60	5	0.60	4.00
0.50	6	0.50	4.50
0.40	7	0.40	4.90
0.30	8	0.30	5.20
0.20	9	0.20	5.40
0.10	10	0.10	5.50

Total worth sums the marginal column downward. Two quarts are worth \$1.90 — not $\$0.90 \times 2$ — and note her spending rose only \$0.80 for a quart worth \$0.90.

Worth, Spending, and Surplus All Come from the Schedule

Price	Quarts/wk	Marginal worth	Total worth	Spending ($P \times Q$)
\$1.00	1	\$1.00	\$1.00	\$1.00
0.90	2	0.90	1.90	1.80
0.80	3	0.80	2.70	2.40
0.70	4	0.70	3.40	2.80
0.60	5	0.60	4.00	3.00
0.50	6	0.50	4.50	3.00
0.40	7	0.40	4.90	2.80
0.30	8	0.30	5.20	2.40
0.20	9	0.20	5.40	1.80
0.10	10	0.10	5.50	1.00

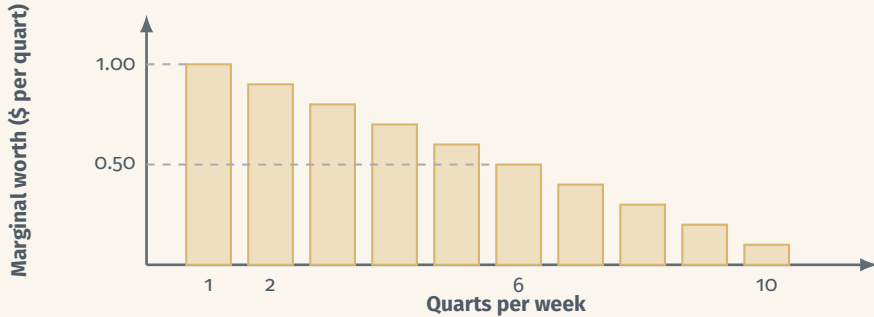
Spending multiplies across: price $\times$ quantity. Watch it rise, peak at \$3.00, and fall — spending is not worth.

Worth, Spending, and Surplus All Come from the Schedule

Price	Quarts/wk	Marginal worth	Total worth	Spending ($P \times Q$)	Surplus
\$1.00	1	\$1.00	\$1.00	\$1.00	\$0.00
0.90	2	0.90	1.90	1.80	0.10
0.80	3	0.80	2.70	2.40	0.30
0.70	4	0.70	3.40	2.80	0.60
0.60	5	0.60	4.00	3.00	1.00
0.50	6	0.50	4.50	3.00	1.50
0.40	7	0.40	4.90	2.80	2.10
0.30	8	0.30	5.20	2.40	2.80
0.20	9	0.20	5.40	1.80	3.60
0.10	10	0.10	5.50	1.00	4.50

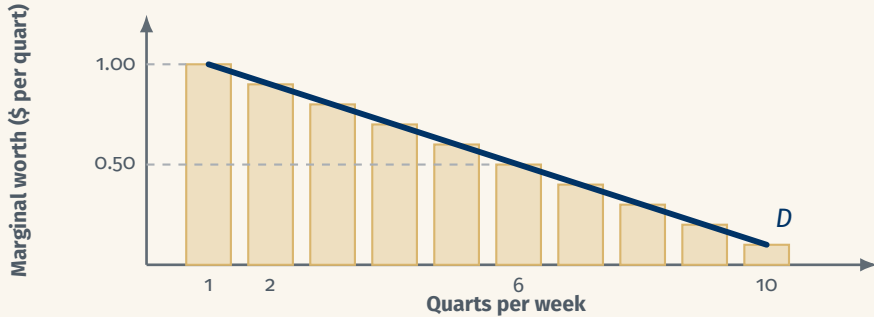
Surplus = total worth – spending. At \$0.50: \$4.50 – \$3.00 = \$1.50 – the gain that makes her buy at all.

Falling Marginal Worth Draws the Demand Curve



This is the table's **marginal worth** column, drawn: each bar is the most she would pay for that successive quart.

Falling Marginal Worth Draws the Demand Curve



Connect the marginal-worth heights: **the resulting demand curve slopes down because later units serve lower-valued uses.**

Same \$3.00 Either Way. So Is the Sixth Quart Worth Nothing?

Price	Quarts	Spent
\$0.60	5	\$3.00
\$0.50	6	\$3.00

She buys one more quart and spends not one cent more. What is that sixth quart worth to her?

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Price	Quarts	Spent
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She buys one more quart and spends not one cent more. What is that sixth quart worth to her?

THE TRAP

“Nothing — spending didn’t change”

A total read where a margin belongs.

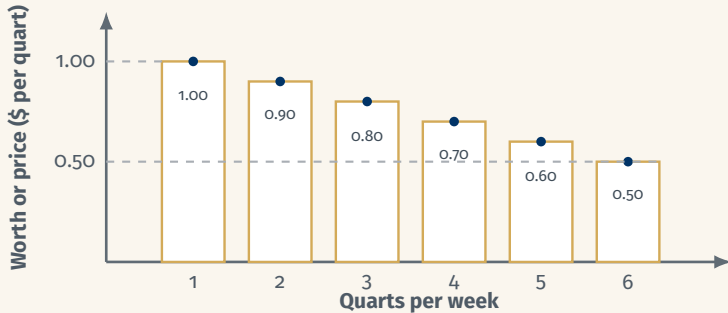
THE ANSWER

\$0.50

The price at which she was just willing to take it.

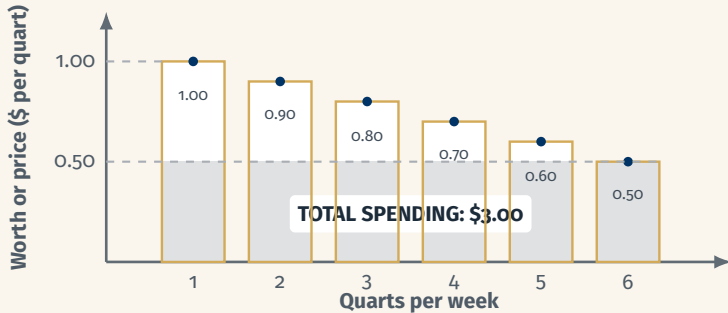
Equal totals are an arithmetic coincidence.

The Buyer Keeps the Gap: Worth Minus Spending



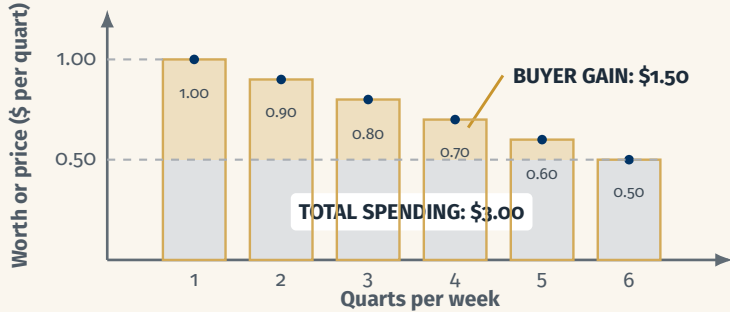
The table's last three columns, drawn: six marginal worths — together, the \$4.50 of total worth.

The Buyer Keeps the Gap: Worth Minus Spending



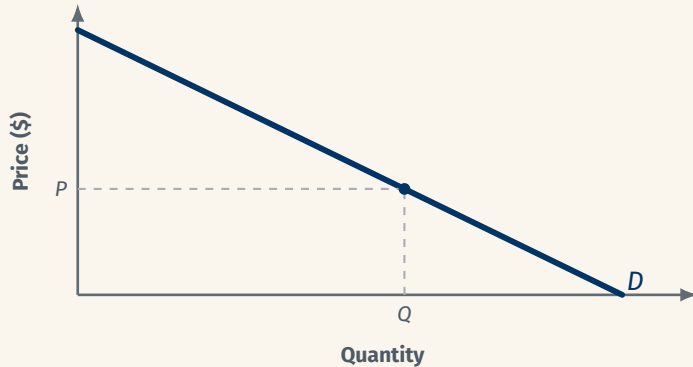
At \$0.50 each, six quarts cost exactly \$3.00.

The Buyer Keeps the Gap: Worth Minus Spending



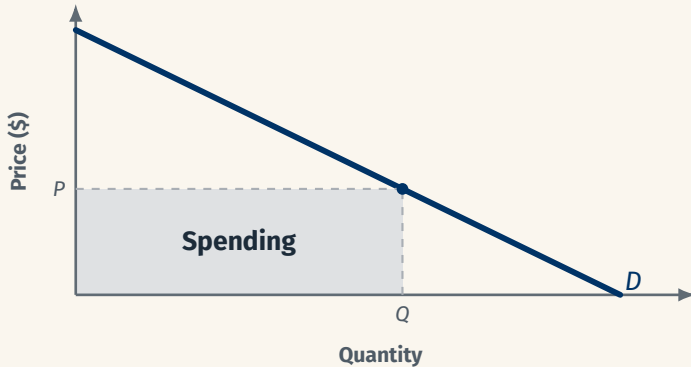
Consumer surplus = total personal worth – total spending: \$4.50 – \$3.00 = \$1.50.

Drawn Smooth, the Surplus Is the Area Above the Price



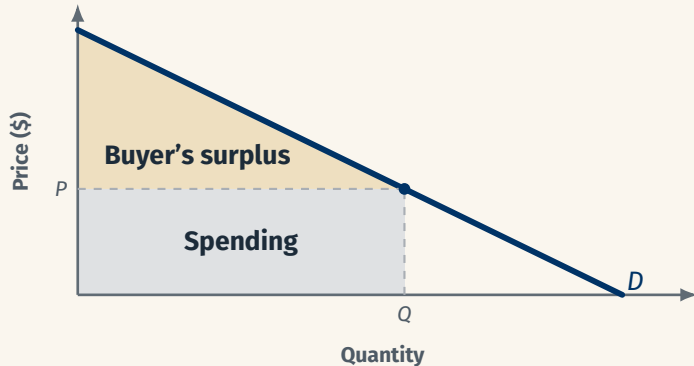
The milk bars, smoothed into a curve: at price P the buyer stops at Q .

Drawn Smooth, the Surplus Is the Area Above the Price



Spending is the rectangle: price times quantity.

Drawn Smooth, the Surplus Is the Area Above the Price



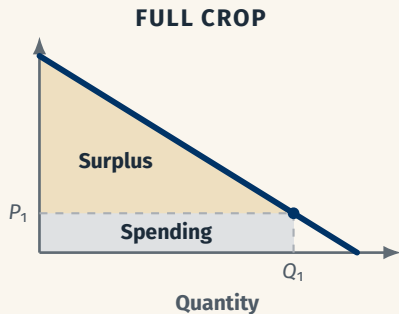
The surplus is the area between the curve and the price — worth above what was paid, on every unit before the last.

A Frost Kills Half the Oranges. Revenue Rises. Anyone Harmed?

The smaller harvest brings more total dollars. So nobody was harmed?

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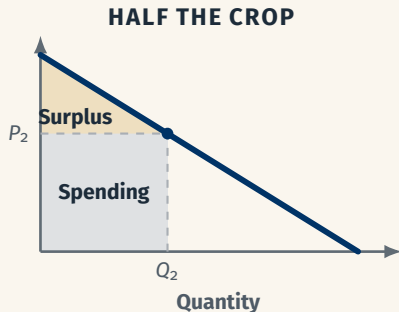
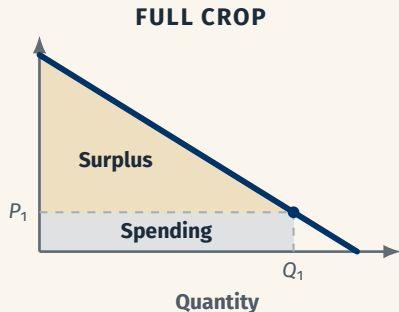
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At the full crop, buyers pay the low price P_1 and keep a large surplus.

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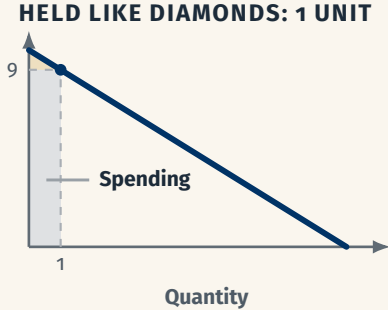
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After the frost the spending rectangle grows — and the surplus shrinks. Revenue is not welfare: buyers are worse off.

Why Is Water Cheap and Diamonds Expensive?

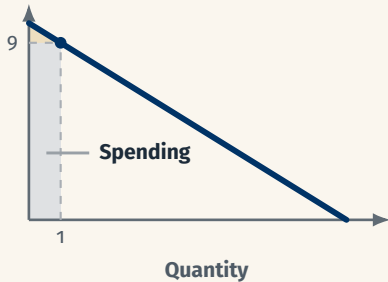
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Last unit worth \$9 — the price is tall

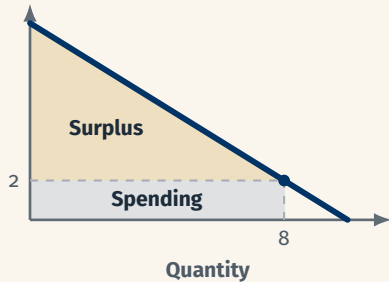
Why Is Water Cheap and Diamonds Expensive?

HELD LIKE DIAMONDS: 1 UNIT



Last unit worth **\$9** — the price is tall
Price tracks the last unit, not the total.

HELD LIKE WATER: 8 UNITS



Last unit worth **\$2** — but look at the surplus

04

Rates, Prices, and “Need”

Back to the question we opened with — now with tools to answer it.

A Demand Claim Must Say What the Quantity Means

RATE OF USE

**The quarts you drink
per week**

A flow — it needs a time period. Our milk schedule is this kind.

ADDITION TO STOCK

**The half-dozen eggs
bought Saturday**

A purchase to top up what is on hand.

STOCK TO OWN

**The quarts kept in the
refrigerator**

Rackets, cars, and rings are demanded this way.

“This buyer demands six quarts.” What is missing?

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“This buyer demands six quarts.” What is missing?

Which of the three it is — and for a rate, the period: six quarts *per what?*

Prices Are Exchange Rates, Even Without Money

A disaster zone: cash has stopped circulating entirely. Does water still have a price?

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1 water bottle = 3 granola bars

RELATIVE PRICE

Three bars per bottle

The sacrifice is real even when no dollars change hands.

MONEY PRICE

Dollars per bottle

Money is the common good in which we usually quote the same trade-off.

“Our City Needs More Golf Courses.” What Is Wrong with That?

The report reads: many people do not play as often as they would like, because of the lack of courses.

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THE MISSING WORDS

“As often as they would like” — at what price?

At a low enough price that is true of nearly everything.

THE TEST

Swap the noun

“Our city needs more filet mignon. People do not eat it as often as they would like.”
Same sentence.

A need label names a priority, not a quantity.

Six Moves Turn a Demand Curve Into a Diagnostic Tool

- 01 Name the buyer, the good, and the quantity — rate or stock, per what?
- 02 Own relative price changed? Slide along the curve.
- 03 Relevant condition changed? Shift; say why.
- 04 Read the height: what is the next unit worth?
- 05 Sum worths; subtract spending for buyer surplus.
- 06 Need claim? Ask outcome, amount, and sacrifice.

THURSDAY We know quantity falls when price rises enough. By how much — and who cares?

BEFORE YOU GO

Tell me what worked and what didn't.



Scan for a one-minute anonymous form.